

FUNDAMENTAL PARTICLES

Several hundred different particles have been discovered in particle accelerator experiments.

Particles are classified into two groups - **hadrons** and **leptons**.

- **Hadrons** - can be divided into **baryons** and **mesons**.
- **Leptons** - are elementary particles with no internal structure.
e.g. electrons, neutrinos.

Particles can be classified according to their spin.

- **Fermions** - have a half-integer spin ($1/2$, $3/2$).

Baryons and **Leptons** are fermions.

Fermions obey the **Pauli Exclusion Principle** - any two fermions in the same quantum state cannot have the same quantum number.

- **Bosons** - have an integer spin (0, 1, 2, ...).

Mesons and **photons** are bosons.

Bosons do not follow the exclusion principle.

The following tables show the various properties of the common leptons, mesons and baryons.

Note

Photons are not listed in the table - they are **field particles** and are involved in interactions between particles

Table 9.1 Some particles and their properties

Category	Particle name	Symbol	Anti-particle	Mass (MeV/c ²)	B	L _e	L _μ	L _τ	Lifetime (s)	Spin
Leptons	Electron	e ⁻	e ⁺	0.511	0	+1	0	0	Stable	$\frac{1}{2}$
	Electron-neutrino	ν _e	$\bar{\nu}_e$	<7 eV/c ²	0	+1	0	0	Stable	$\frac{1}{2}$
	Muon	μ ⁻	μ ⁺	105.7	0	0	+1	0	2.20 × 10 ⁻⁶	$\frac{1}{2}$
	Muon-neutrino	ν _μ	$\bar{\nu}_\mu$	<0.3	0	0	+1	0	Stable	$\frac{1}{2}$
	Tau	τ	τ ⁺	1 784	0	0	0	+1	<4 × 10 ⁻¹³	$\frac{1}{2}$
	Tau-neutrino	ν _τ	$\bar{\nu}_\tau$	<30	0	0	0	+1	Stable	$\frac{1}{2}$
Hadrons										
Mesons	Pion	π ⁺	π ⁻	139.6	0	0	0	0	2.60 × 10 ⁻⁸	0
		π ⁰	Self	135.0	0	0	0	0	0.83 × 10 ⁻¹⁶	0
	Kaon	K ⁺	K ⁻	493.7	0	0	0	0	1.24 × 10 ⁻⁸	0
		K _s ⁰	$\bar{K}_s^0$	497.7	0	0	0	0	0.89 × 10 ⁻¹⁰	0
		K _L ⁰	$\bar{K}_L^0$	497.7	0	0		0	5.2 × 10 ⁻⁸	0
	Eta	η	Self	548.8	0	0	0	0	<10 ⁻⁸	0
		η'	Self	958	0	0	0	0	2.2 × 10 ⁻¹⁰	0

Category	Particle name	Symbol	Anti-particle	Mass (MeV/c ²)	B	L _e	L _μ	L _τ	Lifetime (s)	Spin
Baryons	Proton	p	$\bar{p}$	938.3	+1	0	0	0	Stable	$\frac{1}{2}$
	Neutron	n	$\bar{n}$	939.6	+1	0	0	0	614	$\frac{1}{2}$
	Lambda	Λ ⁰	$\bar{\Lambda}^0$	1115.6	+1	0	0	0	2.6 × 10 ⁻¹⁰	$\frac{1}{2}$
	Sigma	Σ ⁺	$\bar{\Sigma}^-$	1189.4	+1	0	0	0	0.80 × 10 ⁻¹⁰	$\frac{1}{2}$
		Σ ⁰	$\bar{\Sigma}^0$	1192.5	+1	0	0	0	6 × 10 ⁻²⁰	$\frac{1}{2}$
		Σ ⁻	$\bar{\Sigma}^+$	1197.3	+1	0	0	0	1.5 × 10 ⁻¹⁰	$\frac{1}{2}$
	Delta	Δ ⁺⁺	$\bar{\Delta}^{--}$	1230	+1	0	0	0	6 × 10 ⁻²⁴	$\frac{3}{2}$
		Δ ⁺	$\bar{\Delta}^-$	1231	+1	0	0	0	6 × 10 ⁻²⁴	$\frac{3}{2}$
		Δ ⁰	$\bar{\Delta}^0$	1232	+1	0	0	0	63 × 10 ⁻²⁴	$\frac{3}{2}$
		Δ ⁻	$\bar{\Delta}^+$	1234	+1	0	0	0	6 × 10 ⁻²⁴	$\frac{3}{2}$
	Xi	Ξ ⁰	$\bar{\Xi}^0$	1315	+1	0	0	0	2.9 × 10 ⁻¹⁰	$\frac{1}{2}$
		Ξ ⁻	$\bar{\Xi}^+$	1321	+1	0	0	0	1.64 × 10 ⁻¹⁰	$\frac{1}{2}$
	Omega	Ω ⁻	Ω ⁺	1672	+1	0	0	0	0.82 × 10 ⁻¹⁰	$\frac{3}{2}$

Summary

Hadrons

- Made up of three quarks.
- Interact via the strong nuclear, electromagnetic, gravitational and weak forces.
- Divided into mesons and baryons

Mesons

- Have integer spins (0, 1, 2, ...).
- Are bosons.
- No stable meson exists - they all decay.
- Meson decay produces electrons, positrons, neutrinos and photons.

Baryons

- Mass equal or greater than a proton.
- Have half-integer spins (1/2, 3/2).
- Protons and neutrons are baryons - protons are the only stable ones.

Leptons

- Don't interact via the strong nuclear force - only the other three.
- All have spin 1/2.
- Have no internal structure and are point-like.

Field particles

- The four fundamental forces are mediated by a field particle or exchange particle.

Table 9.2 Forces and their exchange particles

Force	Exchange particle
Electromagnetic	Photon, γ
Gravitational	Graviton (not yet detected)
Weak	W and Z bosons, W^+ , W^- , Z^0
Strong nuclear (not a fundamental force)	Pions, π^+ , π^- , π^0

Conservation of Charge and Quantum Numbers

The quantum number for **charge** is conserved in particle decays.

- ▲ A physicist proposed that she had observed a collision of a proton and a neutron, and the products were two protons and an antiproton. Is this reaction allowed according to the law of conservation of charge?

2 parts

Thinking

To find out if this reaction is allowed, calculate the total charge present before and after the interaction.

Working

Proton's charge = $+1$

Neutron's charge = 0

Antiproton's charge = -1

Total charge before = $+1 + 0$
= $+1$

Total charge after = $(+1) + (+1) + (-1)$
= $+1$

Thinking

Compare the total charge before and after the interaction. If they are equal then the reaction is allowed; if they are different then the reaction is forbidden.

Working

The total charge before and after is the same. Therefore, this proposed reaction is allowed according to the law of conservation of charge.

Conservation of Baryon Number

Baryons are all composite particles containing combinations of three quarks.

e.g. Protons and neutrons.

All baryons are given a quantum number - $B = +1$.

For antibaryons - $B = -1$.

For all of other particles involved in the interaction or decay - $B = 0$.

▲ A physicist proposed that she had observed a collision of a proton and a neutron and the products were two protons and an antiproton. Is this reaction allowed according to the law of conservation of baryon number? 2 parts

Thinking

The law of conservation of baryon number states that the net number of baryons remains constant in any process. To find out if this reaction is allowed, calculate the total baryon number before and after the interaction.

Working

Proton $B = +1$

Neutron $B = +1$

Antiproton $B = -1$

Total B before = $(+1) + (+1)$
 $= +2$

Total B after = $(+1) + (+1) + (-1)$
 $= +1$

Thinking

Compare the total baryon number before and after the interaction. If they are equal then the reaction is allowed according to the law of conservation of baryon number. If they are different then the reaction is forbidden as baryon number is not conserved.

Working

The total baryon number before the reaction is different to the total baryon number after the reaction. Therefore, this proposed reaction is forbidden according to the law of conservation of baryon number.

Conservation of Lepton Number

Leptons are the fundamental particles shown below.

Charge	Leptons		
	electron	muon	tau
-1	e^-	μ^-	τ^-
	electron neutrino	muon neutrino	tau neutrino
0	ν_e	ν_μ	ν_τ

Lepton number, L , is a quantum number with value of **+1 for leptons**, **-1 for antileptons** and **0 for other particles**.

There are actually **three types of lepton numbers**: one type for the electron and the electron neutrino, L_e ; one type for the muon and the muon neutrino, L_μ ; and one type for the tau and the tau neutrino, L_τ .

Each of these three lepton numbers must be considered and conserved independently. Both must be conserved for the reaction to be allowed.

▲ A physicist proposed that she had observed the decay of a muon (μ^-) into an electron (e^-), an antielectron neutrino ($\bar{\nu}_e$) and a muon neutrino (ν_μ). Is this reaction allowed according to the law of conservation of lepton number?

5 parts

Thinking Identify which types of leptons are involved in the interaction. These need to be considered separately.	Working electron/electron neutrino, L_e: yes muon/muon neutrino, L_μ: yes tau/tau neutrino, L_τ: no
Thinking To find out if this reaction is allowed, calculate the total electron lepton number L_e before and after the interaction.	Working Electron $L_e = +1$ Antielectron neutrino $L_e = -1$ Total before $L_e = 0$ Total after $L_e = (+1) + (-1) = 0$
Thinking Compare the total electron lepton number before and after the interaction. If they are equal then the reaction is allowed. If they are different then the reaction is forbidden.	Working The total electron lepton number before the reaction is the same as the total electron lepton number after the reaction. Therefore, this proposed reaction is allowed according to the law of conservation of electron lepton number.

Thinking To find out if this reaction is allowed, calculate the total muon lepton number L_μ before and after the interaction.	Working Muon $L_\mu = +1$ Muon neutrino $L_\mu = +1$ Total before $L_\mu = +1$ Total after $L_\mu = +1$
Thinking Compare the total muon lepton number before and after the interaction. If they are equal then the reaction is allowed. If they are different then the reaction is forbidden.	Working The total muon lepton number before the reaction is the same as the total muon lepton number after the reaction. Therefore, this proposed reaction is allowed according to the law of conservation of muon lepton number.

INTERACTIONS BETWEEN PARTICLES

Interactions between particles that produce some change and are observed by a particle detector are commonly known as **events**.

Those interactions that result in the formation of new particles are sometimes referred to as **reactions**.

Interactions can involve attractive or repulsive forces, decay or annihilation.

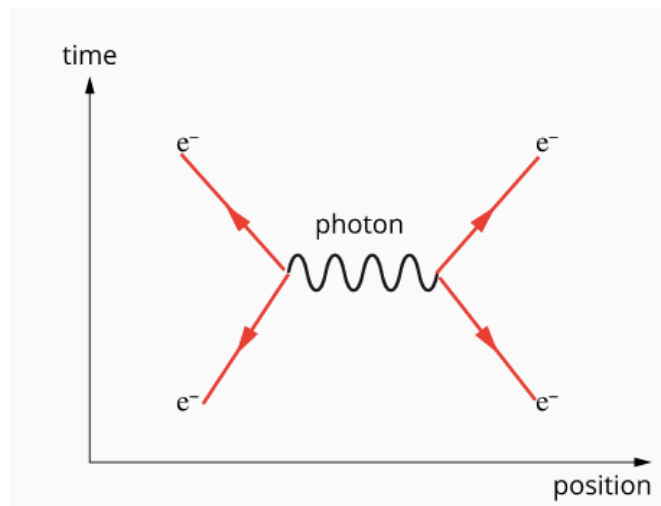
Examples

An example of a **repulsive interaction** involving the electromagnetic force would be the scattering of one electron by another.

This example is illustrated below using a diagram called a **Feynman diagram**.

It shows two electrons approach each other (on the horizontal axis) over time (the vertical axis) and exchange a photon.

This causes each electron to be repelled and therefore change direction.

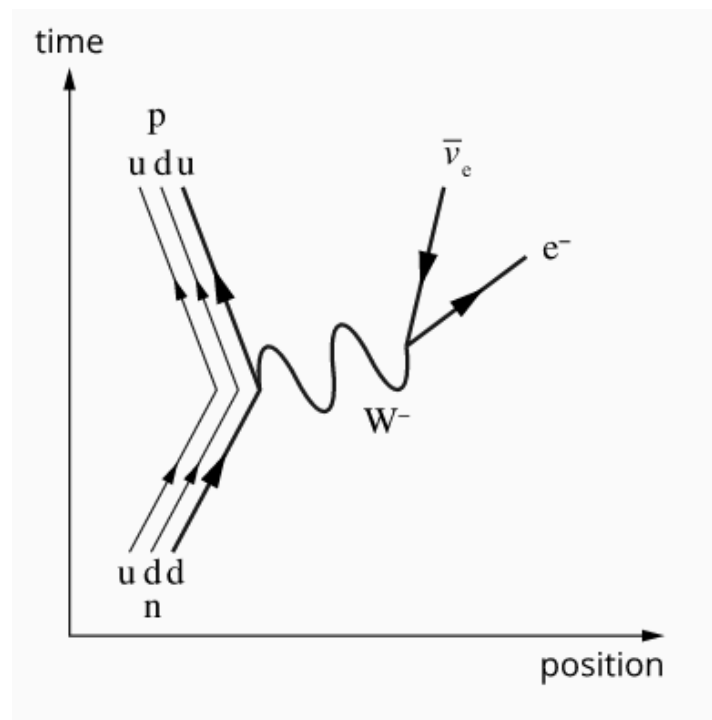


An example of a **decay** is the decay of a neutron into a proton.

This involves the weak nuclear force and is brought about by the conversion of a down quark in the neutron into an up quark.

This requires the emission of a W^- boson, which decays into an antielectron neutrino and an electron.

This is commonly known as a **beta-minus radioactive decay**.



Note

It is important to note that the directions of the arrows in these diagrams do not indicate the direction a particle travels.

e.g. ***An antiparticle moves backwards in time.***

Particles are represented by straight lines - the direction of the arrow represents the direction of time.

For antiparticles - the direction is reversed.

Wiggly lines represent massless exchange particles such as photons.